Volume 05 - Enterprise Data Mesh

Enterprise Google Cloud Data Platform Blueprint Series

Enterprise Data Platform Architecture Office

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1 Document Control

Attribute	Value
Document	Volume 05 - Enterprise Data Mesh
Series	Enterprise Google Cloud Data Platform Blueprint
Version	1.0
Status	Detailed reference architecture
Audience	CDO, CIO, CTO, governance council, domain owners, platform teams, security, analytics, AI and FinOps teams
Scope	Domain-oriented data ownership, data products, federated governance, self-service platform, marketplace, access, operating model and example domains
Baseline scale	Fortune 100 enterprise, 10-50 PB historical data, 50-250 TB daily ingestion, billions of events per day
Primary dependency volumes	Volume 01 Executive Architecture Guide, Volume 02 Landing Zone, Volume 03 Enterprise Data Platform
Implementation note	Product capabilities, quotas and regional availability must be validated against current Google Cloud documentation before production implementation

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24. Implementation Roadmap

25. Validation Checklist

26. Appendices and Reference Implementations

2.1 Executive Summary

This volume defines the enterprise data mesh operating and technical architecture for a global Fortune 100 organization. The data mesh is not treated as a tool deployment or a collection of independent marts. It is a socio-technical architecture in which data is owned by business domains, delivered as governed products, published through a self-service platform, and controlled through federated computational governance. The design is intended to scale to petabytes of data, thousands of pipelines, hundreds of data products, and global regulatory boundaries.

The architecture uses Google Cloud capabilities as the paved road: BigQuery and BigLake for product interfaces, Cloud Storage for open-format persistence, Dataform and Dataflow for standardized product build patterns, Knowledge Catalog for discovery and metadata governance, policy tags and data policies for fine- grained access, Analytics Hub for managed sharing, Cloud Monitoring and Logging for operational telemetry, and Terraform for repeatable domain provisioning. This follows Google Cloud data mesh guidance that treats data products as owned, discoverable, governed and consumable assets developed by the teams that best understand them.

Executive decision	Recommendation	Reason
Adopt data-as-a-product	Mandatory	The enterprise cannot scale analytics and AI if curated data remains project-specific and ownerless.
Use federated governance	Mandatory	Central policy with domain execution avoids both central bottlenecks and uncontrolled decentralization.
Create domain project patterns	Mandatory	A consistent project, dataset and IAM pattern keeps autonomy within security and operational guardrails.
Publish products through marketplace	Mandatory for certified products	A single discovery and access path reduces hidden data copies and improves auditability.
Use contracts and SLOs	Mandatory	Consumers need stable schemas, documented semantics, measurable freshness and support

Executive decision	Recommendation	Reason
		commitments.
Tie cost to product and domain	Mandatory	Data mesh without FinOps becomes decentralized cost sprawl.

The primary implementation risk is organizational rather than technical: domains must accept ownership for product quality, support, cost, lifecycle and documentation. The platform team must therefore provide easy product onboarding, reusable templates, default monitoring, default security controls and automated publication workflows. The governance council must focus on policy, certification and exception management, not manual approval of every dataset.

2.2 Inherited Enterprise Assumptions

This volume inherits the enterprise baseline from the previous volumes and resolves open inputs through explicit planning assumptions. These assumptions are used to produce implementation-ready guidance while leaving final values open for enterprise approval.

Category	Assumption	Architecture implication
Organization	Global Fortune 100 with federated business units	Data mesh will use both central platform teams and domain product teams.
Scale	10-50 PB historical data and 50-250 TB per day	Products require automated metadata, observability and cost controls.
Regions	Multiple Google Cloud regions with residency constraints	Policy tags, catalog entries and data products must be location-aware.
Consumers	BI, AI, applications, regulators and external partners	Products need multiple interfaces: BigQuery, BigLake, APIs, events and Analytics Hub.
Compliance	GDPR, HIPAA, PCI DSS, SOC 2, ISO 27001 and NIST-aligned controls	Data products require classification, lineage, access evidence and retention controls.
Operating model	Central platform plus federated domain teams	Responsibilities must be explicit in RACI and product lifecycle gates.
Security	Zero Trust and least privilege	Access is purpose-based and time-bound for sensitive products.

2.3 Data-Mesh Vision

The enterprise data mesh vision is to make high-value enterprise data discoverable, trusted, secure and reusable across business domains without forcing all data engineering work through a single central delivery queue. The target state is a marketplace of certified data products, each with a clear owner, semantic contract, security classification, SLO, cost owner, lineage trail and support model.

The mesh deliberately separates product ownership from platform ownership. Domains own meaning, quality, prioritization and lifecycle. The platform team owns reusable capabilities, templates, control automation and platform reliability. The governance function defines the rules and makes them machine-enforceable wherever possible.

Outcome	Target state	Success measure
Speed	Domains can publish approved products without building bespoke infrastructure	New standard data product published in less than 4 weeks after discovery.
Trust	Consumers can see owner, lineage, freshness and quality before use	95% of certified products have complete metadata and active quality rules.
Reuse	Products are shared rather than copied into isolated marts	Year-over-year reduction in duplicate curated datasets.
Security	Sensitive products apply policy tags, row filters, masking and audited access	100% of restricted products have automated access evidence.
AI readiness	Certified products can safely support AI and RAG use cases	All AI-approved products have semantic descriptions and usage restrictions.

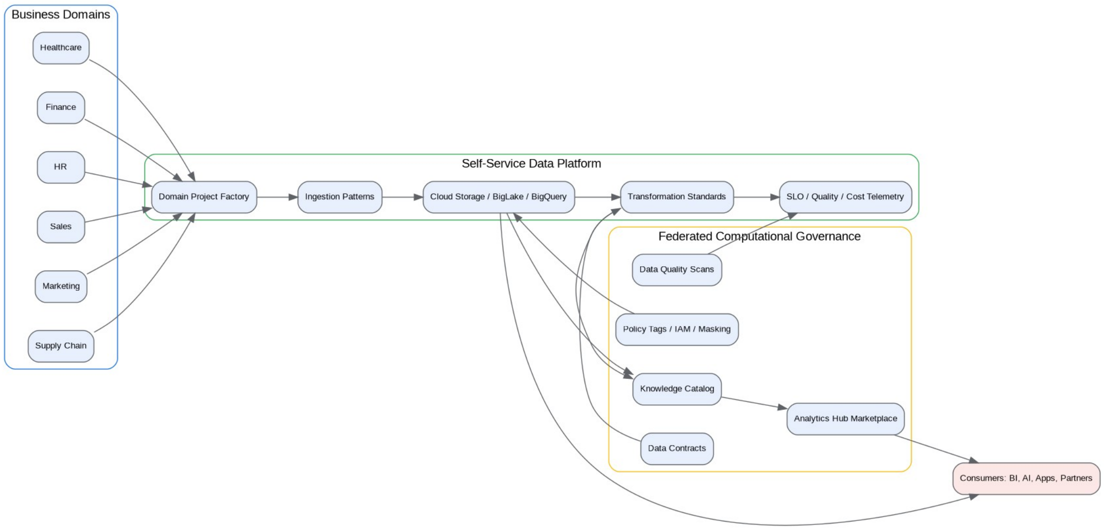
2.4 Data-Mesh Principles

Principle	Implementation meaning
Domain ownership	The domain that understands the business semantics owns the data product, not merely the pipeline. Ownership includes definitions, quality, lifecycle, support and cost.
Data as a product	A dataset becomes a product only when it has a consumer-oriented interface, contract, documentation, quality controls, support model and version policy.
Self-service with guardrails	Domains should not open platform tickets for every standard product. Provisioning, templates, quality checks, security policies and publication paths are automated.

Principle	Implementation meaning
Federated computational governance	Governance policies are centrally defined but enforced through code, metadata, IAM, policy tags, CI/CD checks and catalog workflows.
Interoperable product interfaces	Products can be consumed through BigQuery, BigLake, Analytics Hub, APIs, event streams and semantic models depending on consumer needs.
Security by classification	Every product is classified, and access controls are derived from classification, purpose, region, consumer role and contractual restrictions.
Quality before certification	Products cannot be certified until critical quality, freshness and lineage evidence is available and monitored.
Backward-compatible evolution	Product changes must protect consumers through versioning, compatibility rules, deprecation notices and migration windows.
Observable products	Freshness, quality, availability, usage, cost and incidents are visible at product and domain level.
Cost accountability	Every product is tagged, measured and mapped to domain cost centers. Marketplace consumption supports chargeback or showback.
AI-aware semantics	Products intended for AI use include glossary terms, descriptions, policy metadata, sensitivity, lineage and allowed-use statements.
Avoid domain silos	Domain autonomy is bounded by common platform standards, shared identity, common governance, certified semantics and cross-domain contracts.

2.5 Domain Architecture

Domain architecture uses business capability boundaries rather than application ownership alone. A domain represents a stable area of business accountability with its own source systems, data semantics, product roadmap, stewards and consumers. The same enterprise may contain strategic domains such as customer, product, finance and supply chain, plus operational domains such as observability, reference data and workforce.



Enterprise Data Mesh Reference Architecture

Domain type	Definition	Example	Ownership pattern
Core business domain	Domain aligned to a major value stream or business capability	Customer, Finance, Sales, Supply Chain	Business accountable owner and dedicated data product team.
Supporting domain	Domain that supports enterprise operations	HR, Legal, Procurement	Named product owner with shared platform engineering support.
Reference domain	Domain providing common master or reference entities	Country, Currency, Product Reference	Enterprise stewardship and stricter change control.
Platform domain	Domain that provides reusable platform data and telemetry	Cost, Audit, Observability	Owned by central platform and security teams.
Analytical aggregate domain	Cross-domain product assembled for enterprise outcomes	Customer 360, Profitability, Demand Forecast	Joint ownership with lead domain and governance council approval.

Domain decomposition must avoid mirroring every source application. A single domain can consume many applications, and one application can feed several domains. The deciding questions are: who understands the semantics, who is accountable for quality, who can approve changes, and who funds the product roadmap.

2.6 Domain Decomposition

Decomposition criterion	Guidance	Anti-pattern
Business capability	Start with stable business capabilities and value streams.	Using system names such as SAP or Salesforce as domains.
Data ownership	Assign domains where a business owner can be accountable for meaning and quality.	Assigning ownership to a central data engineering queue.
Consumer cohesion	Group products consumed together by a coherent audience.	Splitting highly related attributes into unrelated products.
Regulatory boundary	Separate domains where regulatory controls or residency rules differ.	Mixing restricted health, HR and marketing data without control boundaries.
Change cadence	Separate products with materially different release cadences and risk profiles.	Forcing real-time operational products into monthly finance governance.
Cost accountability	Align products with budget owners.	Publishing products without accountable cost centers.

A practical decomposition workshop should map capabilities, data sources, critical entities, consumers, regulatory concerns and current pain points. The output is a domain candidate map, not a final organization chart. Domains mature through pilot products and governance evidence.

2.7 Domain Ownership

Data mesh succeeds only when ownership is explicit and enforceable. Ownership is not a title in a catalog. It is a commitment to product reliability, quality, change management and support.

Role	Accountability
Domain data owner	Accountable for product purpose, business definitions, funding, priorities, risk acceptance and consumer commitments.
Data steward	Responsible for glossary, metadata completeness, quality rule interpretation, issue triage and certification evidence.
Data product manager	Owns roadmap, demand intake, interface choices, adoption, communication and lifecycle.
Data product engineer	Builds pipelines, tables, tests, contracts, deployment automation and monitoring.
Analytics engineer	Builds semantic models, certified measures,

Role	Accountability
	transformations and BI-friendly interfaces.
Platform product team	Owns templates, landing zone integration, CI/CD, catalog integration, marketplace workflows and paved-road standards.
Security architect	Approves classification patterns, restricted access patterns and exception handling.
FinOps lead	Defines cost allocation, product unit economics and optimization cadence.

2.8 Data-Product Ownership

A data product is a durable, consumer-oriented data asset. It can be a BigQuery table, an authorized view, a BigLake table, a Looker semantic model, an Analytics Hub listing, an API, an event stream or a set of coordinated interfaces. Product ownership is broader than dataset ownership because it includes consumer commitments and lifecycle management.

Ownership dimension	Required evidence
Product identity	Product name, domain, owner, steward, support channel and version.
Business semantics	Definitions, glossary terms, grain, allowed-use statement and known limitations.
Technical interface	Dataset, view, API, event topic, schema and endpoint documentation.
Quality	Rules, thresholds, scorecards, exceptions and remediation owner.
Security	Classification, policy tags, row filters, masking, encryption and access approvals.
Operations	Freshness SLO, availability SLO, incident runbook and escalation path.
Cost	Cost center, labels, consumer charging model and optimization review cadence.

2.9 Central-Platform Responsibilities

Central platform responsibility	Description	Not responsible for
Paved-road provisioning	Reusable project, dataset, IAM, CI/CD, monitoring and catalog templates.	Business prioritization of domain products.
Security guardrails	Baseline IAM, VPC SC integration, policy tags, key management and logging.	Approving every low-risk consumer manually.
Metadata integration	Common product	Writing domain business

Central platform responsibility	Description	Not responsible for
	registration and catalog automation.	definitions.
Quality framework	Reusable quality rule engines, scorecard templates and alerting.	Resolving domain source-data defects.
Marketplace tooling	Analytics Hub, discovery workflows and access request automation.	Owning consumer relationships for every product.
SRE and platform health	Shared platform availability and incident patterns.	Domain-specific freshness and semantic correctness.
FinOps enablement	Labels, dashboards, budget alerts and reservation guidance.	Domain product cost accountability.

2.10 Domain-Team Responsibilities

Domain-team responsibility	Description	Evidence
Product roadmap	Prioritize products based on business value and consumer demand.	Quarterly product plan.
Source understanding	Document source semantics, quality gaps and lineage.	Source-to-product mapping and glossary.
Contract management	Publish and version schemas, quality rules and compatibility policies.	Approved product contract.
Build and operate	Implement pipelines, transformations, monitoring and runbooks.	Deployment logs and runbook links.
Quality remediation	Investigate failed rules and fix source or transformation issues.	Quality incidents and closure evidence.
Consumer enablement	Provide documentation, onboarding and change notices.	Usage guide and communication history.
Cost management	Review product consumption, storage and compute spend.	Monthly cost dashboard.

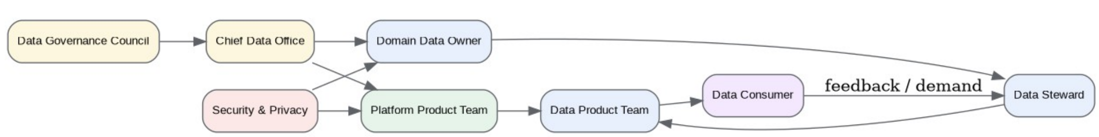
2.11 Self-Service Data Platform

The self-service data platform is the capability layer that makes domain autonomy practical. It hides repeated infrastructure tasks behind approved templates and enables product teams to deliver secure products without bypassing controls. The platform should be product-managed, versioned and measured like any other enterprise platform.

Capability	Self-service feature	Control embedded
Domain onboarding	Project factory creates dev, test and prod domain projects.	Folder placement, billing labels, APIs, IAM groups and logging.
Product skeleton	Repository template creates contract, SQL, tests, documentation and monitoring files.	Naming conventions, schema review and CI quality gates.
Secure storage	Standard datasets and buckets are provisioned by layer and classification.	CMEK options, retention, policy tags and VPC SC placement.
Pipeline templates	Reusable batch, CDC and streaming patterns.	Retry, dead-letter, quality and observability defaults.
Catalog registration	Metadata and glossary templates integrate with Knowledge Catalog.	Mandatory owner, steward, classification and support metadata.
Marketplace publication	Workflow publishes certified products to Analytics Hub.	Certification gate and approval evidence.
Access workflow	Consumers request access through approved groups or listings.	Purpose, classification and recertification controls.

2.12 Federated Computational Governance

Federated computational governance means policies are expressed in a form that can be evaluated by systems, not just described in a policy document. The governance council defines the rules; domains execute within those rules; the platform automates checks and captures evidence.



Federated Governance Operating Model

Governance rule	Computational enforcement
Every product has an owner	CI/CD blocks publication if owner metadata is absent.
Restricted columns are tagged	Schema deployment validates policy tags for sensitive fields.
Quality thresholds are monitored	Data quality scans and assertions publish

Governance rule	Computational enforcement
	scorecards.
Breaking changes are controlled	Contract compatibility checks fail incompatible releases without approved major version.
Products have SLOs	Monitoring policies are created from contract metadata.
Costs are allocated	Required labels and billing export validation are part of product deployment.
Access is purposeful	Access workflow records requester, purpose, approver, expiry and classification.

2.13 Data-Product Lifecycle

The product lifecycle is a managed flow from demand to retirement. Each gate produces evidence that is useful to consumers, governance, security and operations. The lifecycle should be automated through repository templates, CI/CD, catalog APIs, quality scans and access workflows.

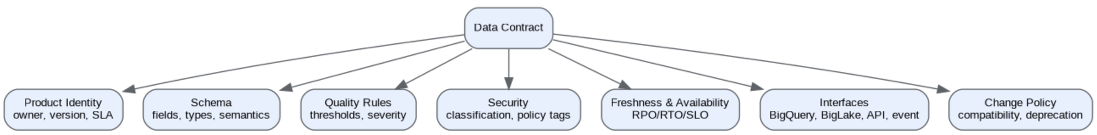
Data Product Lifecycle

Stage	Activities	Exit criteria
Demand intake	Capture use case, consumers, value, data sources and risk.	Product candidate has sponsor, owner and priority.
Design	Define grain, schema, interface, SLOs, classification and cost model.	Contract approved by owner, steward and architecture reviewer.
Build	Implement ingestion, transformation, tests, documentation and monitoring.	All automated tests pass in non-production.
Validate	Run quality, security, lineage, performance and cost checks.	Certification evidence produced.
Publish	Register metadata and expose through approved interface or Analytics Hub.	Consumers can discover and request access.
Operate	Monitor freshness, quality, usage, incidents and cost.	SLO dashboard and runbook active.
Evolve	Release compatible changes or new major version.	Consumers notified and migration plan created.
Deprecate	Mark replacement, freeze changes and schedule	No active consumers or

Stage	Activities	Exit criteria
	retirement.	exception approved.

2.14 Data Contracts

A data contract is the enforceable promise between a producer and consumers. It is not only a schema. It includes semantics, ownership, quality, security, freshness, interface, compatibility, change and support obligations. Contracts should be stored in source control, deployed through CI/CD, and reflected in the catalog.



Data Contract Reference Structure

Contract element	Required content	Validation method
Identity	Product name, domain, owner, steward, support and version.	Contract linter checks mandatory fields.
Schema	Field names, types, descriptions, nullability, primary keys and grain.	Schema compatibility test.
Semantics	Business definitions, glossary terms and permitted use.	Steward review and catalog sync.
Quality	Rules, thresholds, severity and remediation owner.	Dataform assertions or quality scans.
Security	Classification, policy tags, masking, row filters and access conditions.	Policy validation and security review.
Freshness	Expected update frequency, latency SLO and incident severity.	Monitoring policy generated from contract.
Compatibility	Allowed additive changes, breaking-change process and deprecation period.	CI/CD compatibility gate.
Cost	Cost center, chargeback model and consumer impact.	Billing labels and cost dashboard.

2.15 Data-Product Interfaces

Interface type	Use case	Google Cloud pattern	Design controls
BigQuery table	Internal analytical consumption with direct SQL access.	Certified table in product dataset.	Dataset IAM, policy tags, row policies and monitoring.
Authorized view	Filtered or transformed access to sensitive data.	View in sharing dataset authorized to source.	No direct source access for consumer.
Analytics Hub listing	Managed sharing inside or across organizational boundaries.	Exchange and listing for certified product.	Subscription workflow, documentation and audit.
BigLake table	Open-format lakehouse data queried through BigQuery.	BigLake table over Cloud Storage objects.	Fine-grained access without bucket read permissions.
Looker semantic model	Governed measures and dashboard consistency.	LookML model over certified data.	Metric certification and role-based access.
API	Operational or application consumption.	Cloud Run or Apigee backed by product data.	Authentication, quota, audit and data minimization.
Event stream	Near-real-time product changes.	Pub/Sub topic with versioned schema.	Schema evolution, dead-lettering and replay policy.

2.16 Data-Product SLOs

SLO category	Example target	Measurement	Owner
Freshness	95% of product updates available within 15 minutes for Tier 1 products.	Max source event time to product availability.	Domain product team.
Availability	99.9% monthly availability for certified product interface.	Successful query/API/listing availability.	Domain plus platform.
Correctness	Critical quality rules pass at 99.5% or approved exception.	Quality scorecard and rule results.	Data steward.
Completeness	Required source coverage meets published threshold.	Source reconciliation reports.	Data product engineer.
Support	Priority 1 incident	Incident system	Product owner.

SLO category	Example target	Measurement	Owner
	response within 30 minutes.	timestamps.	
Change notice	Breaking change notice at least 90 days before retirement.	Consumer communication log.	Data product manager.

2.17 Data-Product Quality

Quality dimension	Rule example	Certification threshold
Completeness	customer_id is not null.	100% for primary entity products.
Uniqueness	product key is unique for effective timestamp.	100% for keys.
Validity	status is in approved reference set.	99.9% unless exception documented.
Consistency	region_code matches residency rules.	100% for restricted products.
Timeliness	latest partition loaded within SLO.	95-99% by product tier.
Accuracy	financial totals reconcile with source ledger.	100% for regulatory products.
Referential integrity	foreign keys match certified reference products.	99.9% or remediation plan.
Drift	schema and distribution changes are detected.	Alert on unexpected material drift.

2.18 Data-Product Security

Security control	Data mesh implementation
Classification	Product and field classification stored in contract and catalog.
Policy tags	Sensitive columns use taxonomy-aligned policy tags.
Dynamic masking	PII and restricted attributes masked by default for broad consumer groups.
Row access policies	Regional, tenant, legal entity and purpose restrictions enforced in BigQuery.
Authorized views	Consumers access filtered views where direct table access is not justified.
Analytics Hub controls	Listings expose only approved products with documented usage terms.

Security control	Data mesh implementation
IAM groups	Access assigned to groups, not individual users, except temporary break-glass.
Audit evidence	Access requests, query logs and product changes are retained for control testing.
VPC SC alignment	Restricted products remain inside approved service perimeters where applicable.

2.19 Data-Product Versioning

Change type	Version rule	Consumer impact
Add nullable column	Minor version; compatible.	No migration required.
Add required column	Major version unless defaulted.	Consumer migration required.
Rename column	Major version.	Old version maintained during deprecation period.
Change type	Major version unless safe widening.	Compatibility test required.
Change definition	Minor or major depending on semantic impact.	Consumer notice required.
Change freshness SLO	Contract update and approval.	Consumers must accept new SLO.
Remove product	Deprecation and retirement workflow.	Replacement and migration plan required.

2.20 Data-Product Observability

Observable signal	Metric	Dashboard view
Freshness	Source event time to product availability.	Product SLO dashboard.
Quality	Rule pass rate and failed-record count.	Quality scorecard.
Usage	Queries, consumers, subscriptions and API calls.	Adoption dashboard.
Cost	Storage, compute, pipeline and BI cost by product.	FinOps dashboard.
Reliability	Failures, retries, backlog and error budget.	Operations dashboard.
Security	Access grants, denied attempts and sensitive queries.	Security dashboard.
Lineage	Upstream sources and downstream consumers.	Catalog and lineage view.

2.21 Data-Product Cost Allocation

Product-level cost accountability prevents a mesh from becoming a collection of expensive duplications. Costs are tagged at project, dataset, table, job, pipeline and listing levels wherever possible. Billing export to BigQuery provides the measurement source for showback and chargeback.

Cost area	Allocation method	Optimization practice
Storage	Dataset and table labels mapped to product.	Lifecycle policies, partition pruning and duplicate detection.
BigQuery compute	Reservation assignments, job labels and user group mapping.	Slot baselines, query standards and materialized views.
Pipeline compute	Dataflow, Dataproc and Composer labels.	Autoscaling, right-sizing and schedule optimization.
Marketplace consumption	Consumer subscription or query attribution.	Chargeback by consumer domain for high-cost products.
Logging and monitoring	Product and pipeline labels.	Log exclusions and metric retention tiers.
AI consumption	Model, vector index and product labels.	Token budgets and approved AI product tiers.

2.22 Data Marketplace

The data marketplace is the user-facing expression of the mesh. It should allow consumers to discover products, understand trust indicators, compare alternatives, request access, view documentation, inspect lineage, and understand usage obligations. Analytics Hub is used for managed sharing of BigQuery and BigLake- backed data products, while Knowledge Catalog provides broader metadata, glossary and discovery context.

Marketplace capability	Required behavior
Search	Search by business term, domain, owner, classification, product name and use case.
Trust indicators	Show certification, quality score, freshness, lineage completeness and support tier.
Access request	Trigger controlled approval workflow based on classification and purpose.
Documentation	Expose contract, schema, examples, definitions and limitations.
Subscription	Provision Analytics Hub subscription or IAM binding through approved automation.
Feedback	Allow consumers to report defects, request

Marketplace capability	Required behavior
	enhancements and rate product usefulness.
Lifecycle visibility	Show current, deprecated and replacement products.

2.23 Data Discovery

Discovery begins with catalog completeness. A product should not be discoverable as certified unless it has a meaningful description, glossary alignment, owner, steward, classification, quality status, lineage, version and interface documentation. Knowledge Catalog is the system of record for context, while repository metadata remains the source of deployment truth.

Discovery metadata	Why it matters
Business description	Allows non-engineers to determine relevance.
Glossary terms	Aligns product semantics with enterprise vocabulary.
Owner and steward	Establishes accountability and support route.
Classification	Prevents inappropriate access requests.
Lineage	Shows trust, dependencies and impact.
Quality score	Shows whether the product is fit for purpose.
Example query	Accelerates safe adoption.
Allowed use	Prevents misuse in regulatory or AI-sensitive contexts.

2.24 Analytics Hub

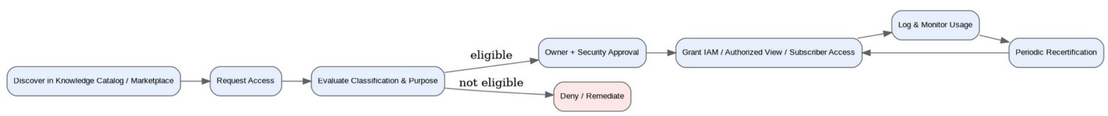
Analytics Hub provides a managed publication and subscription pattern for data products. The mesh uses exchanges for enterprise-wide sharing and domain-specific exchanges for bounded sharing. High-sensitivity products should be exposed through authorized views, filtered listings or request-only workflows rather than broad subscriptions.

Exchange pattern	Use case	Control
Enterprise certified exchange	Approved cross-domain products for broad internal consumption.	Certification required before publication.
Restricted exchange	Sensitive products requiring explicit approval.	Purpose-based approval and recertification.
Partner exchange	Approved partner or external sharing.	Legal agreement, VPC SC review and data minimization.
Domain exchange	Products intended primarily for domain consumers.	Domain owner approval.

Exchange pattern	Use case	Control
AI-ready exchange	Products approved for AI or RAG use.	Responsible AI and sensitive-data controls.

2.25 Access-Request Workflows

Access workflows must balance speed and control. Low-risk certified products can use group-based self-service subscription with automated approval. Restricted or regulated products require owner and security review, explicit purpose, expiry, logging, recertification and revocation controls.



Access Request and Fulfillment Workflow

Step	Description	Evidence
Discover	Consumer finds product in catalog or marketplace.	Search and product view audit.
Request	Consumer states purpose, project, duration and data elements.	Ticket or workflow record.
Classify	Workflow checks product classification and requester group.	Automated policy evaluation.
Approve	Owner, steward, privacy or security approve as required.	Approval record.
Provision	IAM, authorized view or subscription is applied.	Infrastructure or policy change log.
Monitor	Queries, exports and anomalies are monitored.	Audit and security telemetry.
Recertify	Access is reviewed periodically.	Access review evidence.
Revoke	Expired or inappropriate access is removed.	Revocation log.

2.26 Cross-Domain Sharing

Cross-domain sharing is achieved through data contracts, certified product interfaces and marketplace patterns, not through uncontrolled dataset copies. When a product depends on another domain, the consuming domain should treat the upstream product as a contracted dependency with defined SLOs and change notifications.

Sharing pattern	Appropriate use	Risk control
Direct BigQuery access	Trusted internal consumers with low sensitivity.	Dataset IAM and job labels.
Authorized views	Sensitive attributes or regional filters.	No raw table access.
Analytics Hub subscription	Reusable product consumption across teams.	Subscription tracking and marketplace terms.
Product copy	Only where residency, performance or isolation requires it.	Lineage and copy lifecycle controls.
API access	Operational applications and fine-grained request controls.	Authentication, quotas and request audit.
Event subscription	Near-real-time downstream product builds.	Schema versioning and DLQ.

2.27 Data-Product Certification

Certification gate	Mandatory evidence
Ownership	Named owner, steward, product manager and support channel.
Contract	Approved versioned contract with schema and semantics.
Security	Classification, policy tags, masking and access pattern approved.
Quality	Critical rules automated with passing score or approved exception.
Lineage	Upstream and downstream lineage registered or documented.
SLO	Freshness, availability, support and incident severity defined.
Observability	Dashboard and alerting active.
Cost	Labels, cost center and dashboard in place.
Documentation	Catalog record has examples, limitations and permitted uses.
Review	Governance or delegated reviewer approves publication.

2.28 Data-Product Deprecation

Deprecation step	Requirement
Mark deprecated	Catalog and marketplace show deprecated status and replacement.
Notify consumers	Active consumers receive notice through

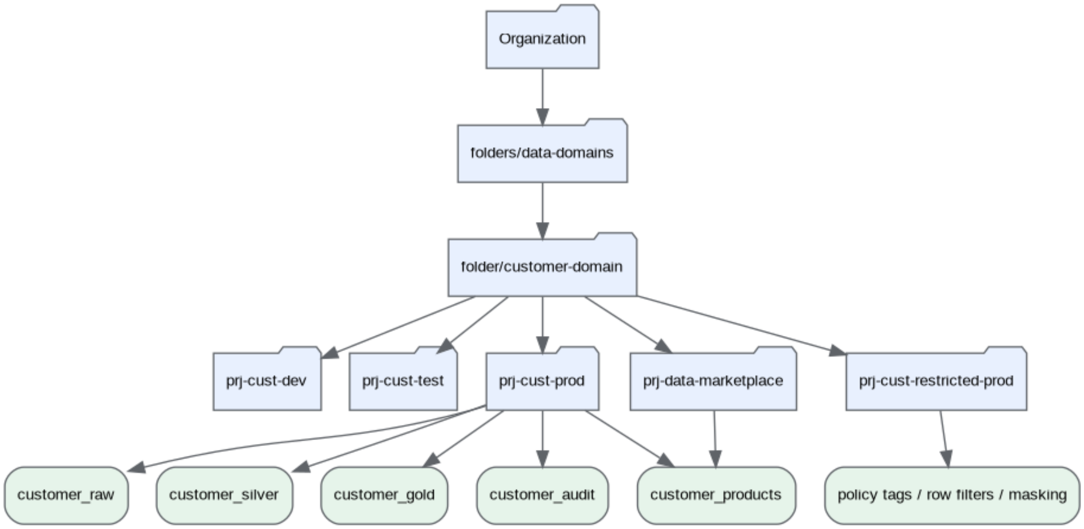
Deprecation step	Requirement
	approved channels.
Freeze breaking changes	Only critical fixes allowed after deprecation.
Migration window	Minimum window depends on tier, typically 90-180 days for certified products.
Monitor usage	Usage dashboard confirms consumer migration.
Remove access	Subscriptions and IAM are revoked after retirement date.
Archive evidence	Contract, lineage, quality and access evidence retained.

2.29 Governance Council, Owners, Stewards and Platform Team

Body or role	Primary purpose	Decision rights
Data Governance Council	Set policies, approve standards, resolve cross-domain conflicts.	Enterprise policy, certification criteria and exception acceptance.
Chief Data Office	Own data strategy, maturity, adoption and governance outcomes.	Prioritization, funding model and stewardship model.
Domain Data Owner	Accountable for domain products and risk.	Product approval, quality exceptions and consumer commitments.
Data Steward	Manage definitions, metadata, quality and certification evidence.	Glossary terms and quality rule interpretation.
Platform Product Team	Build and operate self-service platform capabilities.	Template standards, automation releases and platform roadmap.
Security and Privacy	Approve sensitive-data access patterns and control mappings.	Restricted access, masking standards and exception review.
FinOps Council	Review product cost and optimization.	Budget alerts, chargeback rules and commitment planning.

2.30 Domain Platform Patterns

The domain platform pattern provides a repeatable physical model for domain autonomy. Each domain receives separate non-production and production projects, standardized datasets, controlled service accounts, catalog integration and pipeline templates. Restricted products may be isolated in separate projects or datasets depending on classification and service perimeter design.



Domain Project and Dataset Pattern

Project type	Purpose	Typical resources
domain-dev	Development and unit testing.	Datasets, buckets, Dataform workspace, test topics, temporary compute.
domain-test	Integration testing and certification evidence.	Representative data, CI tests, validation scans.
domain-prod	Production products and pipelines.	Certified datasets, scheduled pipelines, monitoring.
domain-restricted-prod	Highly sensitive products requiring stricter boundaries.	Restricted datasets, CMEK, VPC SC placement, limited IAM.
marketplace	Exchange, listings and authorized sharing views.	Analytics Hub exchanges, sharing datasets.
domain-sandbox	Controlled experimentation with synthetic or de-identified data.	Quota-limited datasets and notebooks.

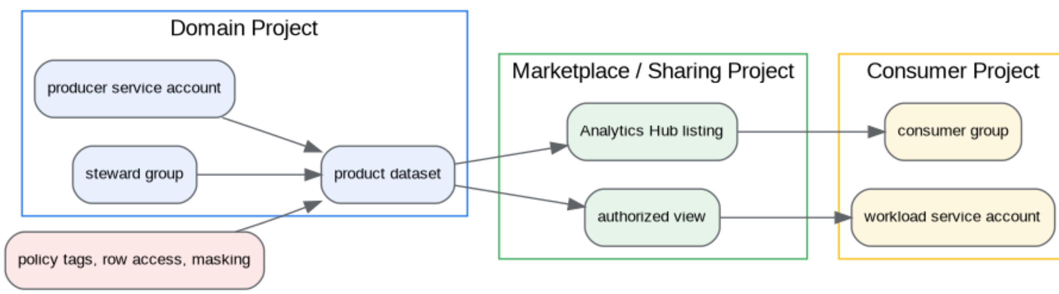
2.31 Domain Dataset Structure

Dataset pattern	Purpose	Access rule
_raw	Immutable source-aligned records.	Producer and engineering access only.
_silver	Standardized, cleaned and conformed records.	Domain engineering and approved dependencies.
_gold	Domain-consumption models.	Approved domain consumers.
_products	Certified products for broad	Published through views,

Dataset pattern	Purpose	Access rule
	sharing.	listings or controlled IAM.
_restricted	Sensitive product data.	Purpose-based access and stronger controls.
_quality	Quality results, rule outputs and scorecards.	Steward and platform operations access.
_audit	Product usage, access and operational evidence.	Domain, security and audit access.

2.32 Data-Mesh IAM Model

The IAM model uses groups for humans, dedicated service accounts for workloads, and product-specific access bindings. Direct user-level grants are avoided except for time-bound break-glass events. Access to products is based on consumer role, purpose, classification, residency and product interface.



Data Mesh IAM and Trust Boundaries

Principal	Grant pattern	Notes
Domain engineers	Project-level development roles in dev/test, limited production deployment role.	Production changes through CI/CD.
Data stewards	Metadata and quality result access; limited data access based on need.	Steward does not automatically get restricted data.
Product pipelines	Dedicated service account per product or product family.	Least privilege to source, target and quality datasets.
Consumers	Group-based access to views/listings; no raw access by default.	Access recertified periodically.
Security reviewers	Read metadata, audit logs and policy evidence.	Restricted data access only when required and approved.
Break-glass	Time-bound elevated group with mandatory review.	All actions logged and reviewed.

2.33 Data-Mesh Network Model

The network model inherits the landing-zone Shared VPC and private access strategy. Data mesh products do not create independent network islands. Domain projects attach to approved host projects, use Private Google Access and Private Service Connect where appropriate, and follow service perimeter boundaries for restricted data. Cross-domain consumption should occur through product interfaces rather than direct private network paths between arbitrary projects.

Network requirement	Pattern
Domain isolation	Separate projects and IAM boundaries rather than separate bespoke VPCs for every product.
Private access	Use private paths to Google APIs for production data movement.
Restricted perimeter	Place restricted products and pipelines in defined VPC SC perimeters where supported.
Hybrid sources	Route source ingestion through approved interconnect or VPN patterns defined in landing zone.
Cross-domain consumption	Prefer BigQuery, Analytics Hub, authorized views and APIs over lateral network access.
Partner sharing	Use approved external sharing patterns with legal, security and data minimization controls.

2.34 Data-Mesh Governance Model

The governance model defines mandatory policies once and implements them through reusable controls. The goal is not to review every table manually; it is to create automated evidence that shows whether products meet standards. Exceptions are allowed but must be recorded, owned, time-bound and reviewed.

Governance area	Policy	Automation
Ownership	Every certified product has owner and steward.	Contract linter and catalog completeness check.
Classification	Every product and sensitive field is classified.	Schema scanner and policy-tag validation.
Quality	Certified products have active quality rules.	Dataform assertions and Knowledge Catalog data quality scans.
Lineage	Products have source and transformation lineage.	Pipeline metadata and catalog lineage.
Access	Restricted access requires	Workflow and IAM

Governance area	Policy	Automation
	approval and expiry.	automation.
Lifecycle	Deprecated products have replacement and retirement date.	Marketplace status and usage monitoring.
Cost	Products carry domain and cost-center labels.	Billing export validation.

2.35 Data-Mesh Maturity Model

Maturity level	Characteristics	Exit criteria
Level 1 - Ad hoc	Datasets exist but ownership, quality and documentation are inconsistent.	Critical products identified and owners assigned.
Level 2 - Managed	Domain teams publish documented products with manual governance.	Standard contracts, quality rules and access patterns used by pilots.
Level 3 - Standardized	Self-service templates and catalog integration are broadly used.	Most certified products follow common lifecycle and SLO model.
Level 4 - Measured	Product usage, quality, SLOs and cost are measured enterprise-wide.	Quarterly product reviews and FinOps routines active.
Level 5 - Optimized	Governance is largely computational and products support AI safely.	Automated controls, high reuse and low duplicate product rate.

2.36 Example Domain: Healthcare

Manage patient, clinical, encounter, claims, provider and care-management data products for analytics, regulatory reporting and AI-assisted care operations.

Dimension	Design
Domain purpose	Manage patient, clinical, encounter, claims, provider and care-management data products for analytics, regulatory reporting and AI-assisted care operations.
Data sources	EHR, HL7/FHIR feeds, claims systems, lab systems, pharmacy, imaging metadata, patient engagement portals, provider directories.
Domain owner	Chief Medical Information Officer or VP Clinical Data
Data steward	Clinical Data Steward

Dimension	Design
Key data products	Patient 360, Encounter Summary, Claims Analytics, Provider Network, Care Gap Registry
Product consumers	Care management, clinical analytics, regulatory reporting, population health, AI care-assist teams
Security classification	Restricted - PHI and confidential clinical data
SLOs	Tier 1 for care operations; 15 minute to 4 hour freshness depending on source
Access model	Authorized views, policy tags, row filters by region/provider, audit-heavy access, no broad raw access
Cost-allocation model	Chargeback to clinical operations, population health and regulatory reporting cost centers.
Data product	Primary interface
—	—
Patient 360	BigQuery certified table, authorized view or Analytics Hub listing
Encounter Summary	BigQuery certified table, authorized view or Analytics Hub listing
Claims Analytics	BigQuery certified table, authorized view or Analytics Hub listing
Provider Network	BigQuery certified table, authorized view or Analytics Hub listing
Care Gap Registry	BigQuery certified table, authorized view or Analytics Hub listing
Quality rule	Severity
—	—
Patient identifier completeness 100%	Critical
Encounter date validity 99.9%	High
FHIR resource conformance 99%	High
Claims reconciliation 100% for financial reporting	Critical
Project or dataset	Example naming
—	—
Development project	prj-healthcare-dev
Production project	prj-healthcare-prod
Restricted project	prj-healthcare-restricted-prod
Product dataset	healthcare_products

Dimension	Design
Quality dataset	healthcare_quality
Audit dataset	healthcare_audit

2.37 Example Domain: Finance

Provide governed financial, ledger, profitability, treasury, risk and regulatory reporting data products.

Dimension	Design
Domain purpose	Provide governed financial, ledger, profitability, treasury, risk and regulatory reporting data products.
Data sources	ERP, general ledger, subledger, treasury, billing, payments, procurement, tax and risk systems.
Domain owner	Corporate Controller or Finance Data Owner
Data steward	Finance Data Steward
Key data products	General Ledger Gold, Profitability Cube, Revenue Recognition, Expense Analytics, Regulatory Finance Pack
Product consumers	FP&A, accounting, treasury, auditors, regulatory reporting, executive dashboards
Security classification	Confidential to restricted depending on legal entity and regulatory scope
SLOs	Tier 0 for close and regulatory periods; Tier 1 otherwise
Access model	Authorized views by legal entity, finance role and close-period status; strict audit retention
Cost-allocation model	Chargeback to finance shared services and consuming business units.
Data product	Primary interface
—	—
General Ledger Gold	BigQuery certified table, authorized view or Analytics Hub listing
Profitability Cube	BigQuery certified table, authorized view or Analytics Hub listing
Revenue Recognition	BigQuery certified table, authorized view or Analytics Hub listing
Expense Analytics	BigQuery certified table, authorized view or Analytics Hub listing
Regulatory Finance Pack	BigQuery certified table, authorized view or

Dimension	Design
	Analytics Hub listing
Quality rule	Severity
—	—
Debit/credit reconciliation 100%	Critical
Currency code validity 100%	Critical
Close-period immutability enforced	High
Legal-entity mapping 100%	Critical
Project or dataset	Example naming
—	—
Development project	prj-finance-dev
Production project	prj-finance-prod
Restricted project	prj-finance-restricted-prod
Product dataset	finance_products
Quality dataset	finance_quality
Audit dataset	finance_audit

2.38 Example Domain: Human Resources

Provide workforce, position, payroll, skills, learning and organization data products with strict privacy controls.

Dimension	Design
Domain purpose	Provide workforce, position, payroll, skills, learning and organization data products with strict privacy controls.
Data sources	Workday, payroll, learning systems, identity systems, recruiting and performance systems.
Domain owner	Chief Human Resources Officer delegate
Data steward	Workforce Data Steward
Key data products	Workforce Headcount, Organization Hierarchy, Skills Inventory, Attrition Analytics, Payroll Controls
Product consumers	HR analytics, workforce planning, finance, access governance, selected executives
Security classification	Restricted - employee personal and compensation data
SLOs	Tier 1 for identity-impacting workforce products; daily for planning products

Dimension	Design
Access model	Default masked compensation and demographic data; purpose-based access with expiry
Cost-allocation model	HR cost center with chargeback for enterprise planning products.
Data product	Primary interface
—	—
Workforce Headcount	BigQuery certified table, authorized view or Analytics Hub listing
Organization Hierarchy	BigQuery certified table, authorized view or Analytics Hub listing
Skills Inventory	BigQuery certified table, authorized view or Analytics Hub listing
Attrition Analytics	BigQuery certified table, authorized view or Analytics Hub listing
Payroll Controls	BigQuery certified table, authorized view or Analytics Hub listing
Quality rule	Severity
—	—
Employee ID uniqueness 100%	Critical
Manager hierarchy validity 99.9%	High
Compensation fields tagged and masked	High
Termination status freshness daily	High
Project or dataset	Example naming
—	—
Development project	prj-human-resources-dev
Production project	prj-human-resources-prod
Restricted project	prj-human-resources-restricted-prod
Product dataset	human_resources_products
Quality dataset	human_resources_quality
Audit dataset	human_resources_audit

2.39 Example Domain: Sales

Provide opportunity, account, pipeline, quota, booking and territory data products for revenue operations.

Dimension	Design
Domain purpose	Provide opportunity, account, pipeline, quota,

Dimension	Design
	booking and territory data products for revenue operations.
Data sources	CRM, CPQ, order management, partner portal, territory systems, sales forecasting tools.
Domain owner	Chief Revenue Officer delegate
Data steward	Sales Operations Data Steward
Key data products	Pipeline Forecast, Account 360, Bookings Gold, Territory Assignment, Sales Activity Metrics
Product consumers	Sales operations, finance, marketing, executive dashboards, AI forecasting teams
Security classification	Confidential with restricted customer and contract details
SLOs	Hourly for pipeline and bookings; daily for territory snapshots
Access model	Role-based views by region, segment and management hierarchy
Cost-allocation model	Revenue operations cost center and showback to sales regions.
Data product	Primary interface
—	—
Pipeline Forecast	BigQuery certified table, authorized view or Analytics Hub listing
Account 360	BigQuery certified table, authorized view or Analytics Hub listing
Bookings Gold	BigQuery certified table, authorized view or Analytics Hub listing
Territory Assignment	BigQuery certified table, authorized view or Analytics Hub listing
Sales Activity Metrics	BigQuery certified table, authorized view or Analytics Hub listing
Quality rule	Severity
—	—
Opportunity amount non-negative 100%	Critical
Close date validity 99%	High
Territory assignment completeness 99.5%	High
Bookings reconciliation with finance 100%	Critical
Project or dataset	Example naming

Dimension	Design
—	—
Development project	prj-sales-dev
Production project	prj-sales-prod
Restricted project	prj-sales-restricted-prod
Product dataset	sales_products
Quality dataset	sales_quality
Audit dataset	sales_audit

2.40 Example Domain: Marketing

Provide campaign, consent, audience, engagement and attribution products for growth and personalization.

Dimension	Design
Domain purpose	Provide campaign, consent, audience, engagement and attribution products for growth and personalization.
Data sources	Marketing automation, web analytics, consent management, ad platforms, CRM, customer data platform.
Domain owner	Chief Marketing Officer delegate
Data steward	Marketing Data Steward
Key data products	Campaign Performance, Audience Segments, Consent Status, Attribution Model, Lead Funnel
Product consumers	Marketing analytics, sales, digital teams, AI personalization, privacy operations
Security classification	Confidential; restricted where PII, consent or tracking identifiers are used
SLOs	Near real time for consent; hourly or daily for analytics
Access model	Consent-aware views, masking of direct identifiers, strict allowed-use statements
Cost-allocation model	Marketing operations with product-level showback for activation workloads.
Data product	Primary interface
—	—
Campaign Performance	BigQuery certified table, authorized view or Analytics Hub listing
Audience Segments	BigQuery certified table, authorized view or

Dimension	Design
	Analytics Hub listing
Consent Status	BigQuery certified table, authorized view or Analytics Hub listing
Attribution Model	BigQuery certified table, authorized view or Analytics Hub listing
Lead Funnel	BigQuery certified table, authorized view or Analytics Hub listing
Quality rule	Severity
—	—
Consent status freshness under 15 minutes	Critical
Campaign ID validity 99.5%	High
No activation without consent rule	Critical
Attribution completeness threshold by channel	High
Project or dataset	Example naming
—	—
Development project	prj-marketing-dev
Production project	prj-marketing-prod
Restricted project	prj-marketing-restricted-prod
Product dataset	marketing_products
Quality dataset	marketing_quality
Audit dataset	marketing_audit

2.41 Example Domain: Supply Chain

Provide supplier, inventory, logistics, demand, order fulfillment and resilience data products.

Dimension	Design
Domain purpose	Provide supplier, inventory, logistics, demand, order fulfillment and resilience data products.
Data sources	ERP, warehouse management, transportation management, supplier portals, IoT sensors, planning systems.
Domain owner	Chief Supply Chain Officer delegate
Data steward	Supply Chain Data Steward
Key data products	Inventory Position, Supplier Risk, Demand Forecast Inputs, Shipment Visibility, Order Fulfillment

Dimension	Design
Product consumers	Operations, procurement, finance, sales, risk, AI demand planning
Security classification	Confidential with restricted supplier and contractual terms
SLOs	Minutes for shipment visibility; hourly/daily for planning products
Access model	Views by business unit, region and supplier confidentiality; partner sharing by contract
Cost-allocation model	Supply chain operations and consuming planning teams.
Data product	Primary interface
—	—
Inventory Position	BigQuery certified table, authorized view or Analytics Hub listing
Supplier Risk	BigQuery certified table, authorized view or Analytics Hub listing
Demand Forecast Inputs	BigQuery certified table, authorized view or Analytics Hub listing
Shipment Visibility	BigQuery certified table, authorized view or Analytics Hub listing
Order Fulfillment	BigQuery certified table, authorized view or Analytics Hub listing
Quality rule	Severity
—	—
SKU validity 99.9%	High
Inventory balance reconciliation 99.5%	High
Shipment event ordering rules	High
Supplier ID completeness 100%	Critical
Project or dataset	Example naming
—	—
Development project	prj-supply-chain-dev
Production project	prj-supply-chain-prod
Restricted project	prj-supply-chain-restricted-prod
Product dataset	supply_chain_products
Quality dataset	supply_chain_quality
Audit dataset	supply_chain_audit

2.42 Technology and Service Mapping

Capability	Primary Google Cloud services	Selection rationale
Product storage and SQL interface	BigQuery, BigLake, Cloud Storage	Supports native warehouse products and open-format lakehouse products.
Product discovery and context	Knowledge Catalog	Provides enterprise context, metadata, governance, data profiling and quality capabilities.
Managed sharing	Analytics Hub	Supports managed exchanges and listings for data products.
Transformation and product build	Dataform, BigQuery SQL, Dataflow	Supports SQL products, batch/stream pipelines and automated tests.
Security and access	IAM, policy tags, data policies, row access policies, VPC SC	Provides layered least-privilege and fine-grained controls.
Observability	Cloud Monitoring, Cloud Logging, BigQuery INFORMATION_SCHEMA	Provides freshness, quality, reliability, usage and cost signals.
Automation	Terraform, Cloud Build, Artifact Registry	Provides repeatable domain and product provisioning.
AI-ready consumption	Vertex AI, BigQuery, Knowledge Catalog	Allows certified products to support AI and RAG with governed context.

Service	Purpose	Benefits
—	—	—
BigQuery	Analytical data product serving layer.	Serverless SQL, fine-grained security, scalable analytics.
BigLake	Governed open-format data access through BigQuery.	Fine-grained access to Cloud Storage data without direct bucket access.
Cloud Storage	Immutable raw and open-format persistence.	Durable, scalable, lifecycle controls.
Knowledge Catalog	Metadata, context, glossary, profiling, quality and lineage.	Enterprise discovery and AI context.
Analytics Hub	Data product marketplace and managed sharing.	Clear producer-consumer model.
Dataform	SQL product transformations and assertions.	Version-controlled BigQuery workflows.

Capability	Primary Google Cloud services	Selection rationale
Dataflow	Batch and streaming product pipelines.	Apache Beam portability and robust stream patterns.
IAM and policy tags	Access control.	Group-based access and column-level control.

2.43 Architecture Decision Records

2.43.1 MESH-ADR-001: Adopt federated data mesh instead of centralized-only data delivery

Field	Content
Status	Proposed for enterprise adoption
Date	2026-07-11
Problem	Centralized data teams cannot scale product delivery across hundreds of domains.
Alternatives considered	Centralized platform only; decentralized marts; federated mesh.
Decision	Adopt federated mesh with central platform and domain ownership.
Tradeoffs	Requires stronger product ownership and governance automation.
Risks	Organizational adoption, inconsistent execution or control gaps if automation is incomplete.
Mitigations	Executive sponsorship, domain KPIs and automated controls.
Review trigger	Annual review or material change to Google Cloud capabilities, regulation or enterprise operating model.

2.43.2 MESH-ADR-002: Use domain-owned data products as the primary delivery unit

Field	Content
Status	Proposed for enterprise adoption
Date	2026-07-11
Problem	Datasets without product commitments are hard to trust or reuse.
Alternatives considered	Dataset ownership; report ownership; product ownership.
Decision	Certified data products become the standard unit of enterprise data consumption.
Tradeoffs	Adds metadata and SLO obligations.

Field	Content
Risks	Organizational adoption, inconsistent execution or control gaps if automation is incomplete.
Mitigations	Self-service templates and maturity-based adoption.
Review trigger	Annual review or material change to Google Cloud capabilities, regulation or enterprise operating model.

2.43.3 MESH-ADR-003: Use Knowledge Catalog as enterprise metadata and discovery plane

Field	Content
Status	Proposed for enterprise adoption
Date	2026-07-11
Problem	Consumers need a single context layer for products and governance.
Alternatives considered	Spreadsheet inventory; third-party catalog; Knowledge Catalog.
Decision	Use Knowledge Catalog as the Google Cloud-native catalog and governance context layer.
Tradeoffs	Requires integration and completeness discipline.
Risks	Organizational adoption, inconsistent execution or control gaps if automation is incomplete.
Mitigations	CI/CD metadata sync and certification gates.
Review trigger	Annual review or material change to Google Cloud capabilities, regulation or enterprise operating model.

2.43.4 MESH-ADR-004: Use Analytics Hub for managed product sharing

Field	Content
Status	Proposed for enterprise adoption
Date	2026-07-11
Problem	Consumers need auditable subscription patterns.
Alternatives considered	Direct dataset IAM; manual copies; Analytics Hub listings.
Decision	Publish certified products through exchanges and listings where appropriate.
Tradeoffs	Requires marketplace operating model.

Field	Content
Risks	Organizational adoption, inconsistent execution or control gaps if automation is incomplete.
Mitigations	Define exchange types and publication standards.
Review trigger	Annual review or material change to Google Cloud capabilities, regulation or enterprise operating model.

2.43.5 MESH-ADR-005: Use BigQuery and BigLake as standard product interfaces

Field	Content
Status	Proposed for enterprise adoption
Date	2026-07-11
Problem	Products must support warehouse and lakehouse patterns.
Alternatives considered	BigQuery only; object storage only; mixed unmanaged interfaces.
Decision	Use BigQuery for curated products and BigLake for governed open-format products.
Tradeoffs	Requires file layout standards for BigLake.
Risks	Organizational adoption, inconsistent execution or control gaps if automation is incomplete.
Mitigations	Define product interface selection matrix.
Review trigger	Annual review or material change to Google Cloud capabilities, regulation or enterprise operating model.

2.43.6 MESH-ADR-006: Mandate data contracts for certified products

Field	Content
Status	Proposed for enterprise adoption
Date	2026-07-11
Problem	Consumers need predictable schema, semantics and freshness.
Alternatives considered	Informal documentation; schema-only contracts; full contracts.
Decision	Full contracts are mandatory for certification.
Tradeoffs	Increases initial product effort.
Risks	Organizational adoption, inconsistent

Field	Content
	execution or control gaps if automation is incomplete.
Mitigations	Provide contract templates and CI validation.
Review trigger	Annual review or material change to Google Cloud capabilities, regulation or enterprise operating model.

2.43.7 MESH-ADR-007: Use policy tags, row policies and masking for sensitive products

Field	Content
Status	Proposed for enterprise adoption
Date	2026-07-11
Problem	Sensitive products need fine-grained controls.
Alternatives considered	Dataset-only IAM; copies with masked data; fine-grained policies.
Decision	Use fine-grained policies plus authorized interfaces.
Tradeoffs	Requires taxonomy governance.
Risks	Organizational adoption, inconsistent execution or control gaps if automation is incomplete.
Mitigations	Automated classification and schema validation.
Review trigger	Annual review or material change to Google Cloud capabilities, regulation or enterprise operating model.

2.43.8 MESH-ADR-008: Require product-level SLOs and observability

Field	Content
Status	Proposed for enterprise adoption
Date	2026-07-11
Problem	Product trust depends on measurable operations.
Alternatives considered	Best effort; platform-only SLO; product SLO.
Decision	Each certified product has freshness, quality and availability SLOs.
Tradeoffs	Requires monitoring cost and ownership.
Risks	Organizational adoption, inconsistent execution or control gaps if automation is incomplete.

Field	Content
Mitigations	Generate default dashboards from templates.
Review trigger	Annual review or material change to Google Cloud capabilities, regulation or enterprise operating model.

2.43.9 MESH-ADR-009: Adopt domain project factory patterns

Field	Content
Status	Proposed for enterprise adoption
Date	2026-07-11
Problem	Domain autonomy needs repeatable infrastructure.
Alternatives considered	Manual projects; central shared project; project factory.
Decision	Provision standardized projects and datasets through Terraform.
Tradeoffs	Initial module investment.
Risks	Organizational adoption, inconsistent execution or control gaps if automation is incomplete.
Mitigations	Reusable modules and onboarding checklist.
Review trigger	Annual review or material change to Google Cloud capabilities, regulation or enterprise operating model.

2.43.10 MESH-ADR-010: Use group-based access and avoid direct user grants

Field	Content
Status	Proposed for enterprise adoption
Date	2026-07-11
Problem	Auditability and lifecycle require group-based control.
Alternatives considered	User grants; group grants; dynamic access service.
Decision	Use groups and product-specific bindings by default.
Tradeoffs	Group lifecycle must be managed.
Risks	Organizational adoption, inconsistent execution or control gaps if automation is incomplete.
Mitigations	Recertification and access expiry.
Review trigger	Annual review or material change to Google

Field	Content
	Cloud capabilities, regulation or enterprise operating model.

2.43.11 MESH-ADR-011: Implement certification before marketplace publication

Field	Content
Status	Proposed for enterprise adoption
Date	2026-07-11
Problem	Marketplace without trust signals creates risk.
Alternatives considered	Open publication; manual review only; automated certification gate.
Decision	Products must pass certification gates before enterprise publication.
Tradeoffs	Could slow early adoption.
Risks	Organizational adoption, inconsistent execution or control gaps if automation is incomplete.
Mitigations	Define pilot maturity exceptions.
Review trigger	Annual review or material change to Google Cloud capabilities, regulation or enterprise operating model.

2.43.12 MESH-ADR-012: Treat AI-ready products as a stricter certification class

Field	Content
Status	Proposed for enterprise adoption
Date	2026-07-11
Problem	AI use increases privacy and misuse risk.
Alternatives considered	Allow all certified products for AI; block AI; AI certification class.
Decision	Create AI-ready certification with allowed-use and guardrails.
Tradeoffs	More governance work.
Risks	Organizational adoption, inconsistent execution or control gaps if automation is incomplete.
Mitigations	Templates and AI review checklist.
Review trigger	Annual review or material change to Google Cloud capabilities, regulation or enterprise operating model.

2.43.13 MESH-ADR-013: Use chargeback/showback at product and domain level

Field	Content
Status	Proposed for enterprise adoption
Date	2026-07-11
Problem	Cost transparency is required for mesh sustainability.
Alternatives considered	Project-only cost; domain cost; product cost.
Decision	Use product labels and billing export for product-level showback and targeted chargeback.
Tradeoffs	Attribution is imperfect for shared workloads.
Risks	Organizational adoption, inconsistent execution or control gaps if automation is incomplete.
Mitigations	Improve labels and query job tagging.
Review trigger	Annual review or material change to Google Cloud capabilities, regulation or enterprise operating model.

2.43.14 MESH-ADR-014: Use versioned deprecation instead of breaking product changes

Field	Content
Status	Proposed for enterprise adoption
Date	2026-07-11
Problem	Consumers need stability.
Alternatives considered	Immediate breaking changes; endless old versions; planned deprecation.
Decision	Breaking changes require major version and migration window.
Tradeoffs	Maintains multiple versions temporarily.
Risks	Organizational adoption, inconsistent execution or control gaps if automation is incomplete.
Mitigations	Usage monitoring and retirement gates.
Review trigger	Annual review or material change to Google Cloud capabilities, regulation or enterprise operating model.

2.43.15 MESH-ADR-015: Use domain stewards as certification evidence owners

Field	Content
Status	Proposed for enterprise adoption
Date	2026-07-11
Problem	Business meaning cannot be delegated entirely to engineers.
Alternatives considered	Engineer-only review; governance-only review; steward-owned evidence.
Decision	Data stewards own business metadata and quality interpretation.
Tradeoffs	Requires staffing.
Risks	Organizational adoption, inconsistent execution or control gaps if automation is incomplete.
Mitigations	Steward enablement and governance workflows.
Review trigger	Annual review or material change to Google Cloud capabilities, regulation or enterprise operating model.

2.44 Risks and Mitigations

Risk	Impact	Mitigation
Domains do not accept ownership	Mesh becomes a catalog of central-team datasets.	Tie ownership to executive KPIs, product funding and certification gates.
Governance becomes manual bottleneck	Product delivery slows and teams bypass controls.	Automate policy checks and delegate low-risk approvals.
Inconsistent product semantics	Consumers lose trust and duplicate datasets.	Use glossary, steward review and product contracts.
Sensitive data is overexposed	Regulatory and reputational impact.	Classification, policy tags, masking, approvals and audit.
Product cost sprawl	FinOps objectives fail.	Product labels, budgets, query standards and chargeback.
Marketplace fills with low-quality products	Discovery quality collapses.	Certification levels and deprecation discipline.
Breaking changes disrupt consumers	Operational and reporting failures.	Versioning, compatibility checks and migration windows.
AI use misapplies data	Privacy or model-risk	AI-ready certification and

Risk	Impact	Mitigation
	incidents.	allowed-use metadata.

2.45 Implementation Roadmap

Phase	Duration	Key outcomes
0. Mobilize	4-6 weeks	Approve operating model, domains, standards and first pilot products.
1. Foundation	6-10 weeks	Deploy product templates, catalog integration, quality framework and access workflow.
2. Pilot domains	10-16 weeks	Publish first certified products in Customer/Finance or similar high-value domains.
3. Scale domains	3-6 months	Onboard additional domains and exchange patterns.
4. Optimize	6-12 months	Product cost optimization, reuse metrics, AI-ready products and automated governance.
5. Institutionalize	Ongoing	Quarterly product reviews, maturity assessments and continuous platform improvements.

Milestone	Acceptance evidence
—	—
First certified product	Contract, catalog entry, quality scorecard, access workflow and dashboard.
First Analytics Hub listing	Exchange, listing, subscription workflow and audit evidence.
First restricted product	Policy tags, masking, row policies and approval workflow.
First cross-domain product	Documented upstream dependency and SLO.
First AI-ready product	Allowed-use statement, classification and AI review evidence.

2.46 Data Mesh Validation Checklist

Validation item	Pass criteria
Ownership	Every certified product has owner, steward and support route.
Contract	Contract validates through CI/CD.
Security	Sensitive fields classified and protected.
Quality	Critical rules are automated and monitored.
Lineage	Upstream sources and downstream consumers documented.
Marketplace	Published product includes metadata, usage terms and access workflow.
SLO	Freshness, availability and correctness are measurable.
Cost	Product spend is visible in FinOps dashboard.
Operations	Runbook and incident process are available.
Deprecation	Versioning and retirement process exists before publication.

2.47 Appendices and Reference Implementations

The package includes Terraform, BigQuery SQL, Dataform SQLX, diagram sources and workshop checklists. These are reference implementations and must be adapted to organization-specific billing accounts, folders, regions, IAM groups, VPC Service Controls, policy tags and naming conventions.

Artifact	Location	Purpose
Domain Terraform module	terraform/domain_module.tf	Reusable domain project and dataset example.
Analytics Hub Terraform	terraform/analytics_hub_listing.tf	Data marketplace listing example.
BigQuery DDL	sql/data_product_ddl.sql	Certified product table pattern.
Dataform SQLX	dataform/customer_360.sqlx	Incremental product build example with assertions.
Workshop checklist	workshops/data_mesh_design_review_checklist.md	Review guide for domain product design.
Diagrams	diagrams/*	Graphviz, Mermaid, PlantUML, Draw.io, SVG and PNG diagrams.

2.48 References

- Google Cloud Architecture Center - Architecture and functions in a data mesh: https://docs.cloud.google.com/architecture/data-mesh
- Google Cloud Architecture Center - Build data products in a data mesh: https://docs.cloud.google.com/architecture/build-data-products-data-mesh
- Google Cloud Knowledge Catalog overview: https://docs.cloud.google.com/dataplex/docs/introduction
- Knowledge Catalog data profiling: https://docs.cloud.google.com/dataplex/docs/data-profiling-overview
- Knowledge Catalog data quality scans: https://docs.cloud.google.com/dataplex/docs/auto-data-quality-overview
- Knowledge Catalog data lineage: https://docs.cloud.google.com/dataplex/docs/about-data-lineage
- BigQuery row-level security: https://docs.cloud.google.com/bigquery/docs/row-level-security-intro
- BigQuery column-level security: https://docs.cloud.google.com/bigquery/docs/column-level-security-intro
- BigQuery dynamic data masking: https://docs.cloud.google.com/bigquery/docs/column-data-masking-intro
- BigQuery policy tags best practices: https://docs.cloud.google.com/bigquery/docs/best-practices-policy-tags

3 Detailed Data Mesh Implementation Guide

This section expands the core architecture into an implementation guide that domain product teams, platform engineers, governance reviewers and security architects can use during design and delivery. It converts the operating model into concrete artefacts, lifecycle gates, role assignments, metadata standards, contract content, product templates, controls, acceptance tests and review cadences. The intent is to remove ambiguity: a team should know when a dataset is not yet a product, when a product is not yet certifiable, and when a published product must be remediated, versioned or retired.

3.1 End-to-End Operating Process

Operating step	Implementation detail
Demand qualification	The product manager captures the consumer problem, expected decisions, regulatory sensitivity, target consumers, existing duplicate products, estimated value, expected frequency, consumer latency requirements and criticality. This prevents the mesh from filling with products that nobody consumes or products that duplicate existing certified assets.

Operating step	Implementation detail
Domain approval	The domain data owner confirms that the candidate product belongs in the domain, has a funding path, and has an accountable product team. If ownership is unclear, the governance council resolves domain placement before engineering begins.
Contract-first design	The team writes the contract before implementation. The contract defines grain, fields, semantic definitions, security classification, freshness, quality rules, compatibility policy and support model. Contract-first delivery prevents pipelines from shaping product semantics by accident.
Security and privacy design	Restricted fields, policy tags, masking, row filters, authorized views, allowed-use statements, residency rules and access workflow requirements are defined during design. Security is not deferred to the publication phase.
Build and test	The product is implemented using approved pipeline templates. Dataform assertions, schema tests, reconciliation tests, policy checks and unit tests run in development and test environments. Changes are reviewed through pull requests and deployed through CI/CD.
Certification review	The steward, owner, platform reviewer and security reviewer evaluate evidence. The product is certified only when mandatory controls pass or a time-bound exception is recorded.
Publication and onboarding	The product is published into the marketplace, consumers are onboarded through the access workflow, examples are provided, and usage tracking begins.
Operate and improve	The team monitors SLOs, quality, incidents, cost, consumer feedback and adoption. The product roadmap is adjusted based on measurable value, not only producer priorities.

3.2 Standard Data Product Template

The following template is mandatory for every certified product. It may be implemented as YAML, JSON, repository metadata, catalog metadata or a combination of these, but the content must remain consistent across the repository, catalog and marketplace listing.

Template section	Mandatory fields	Example
Identity	product_id, display_name, domain, version, status, tier, owner, steward, support_group	customer.customer_360.v1
Business context	purpose, business process, consumers, decisions supported, limitations, allowed use	Used for service segmentation and care planning; not approved for credit decisions.
Semantics	grain, glossary_terms, entity definitions, metric definitions, reference data dependencies	One row per active customer profile per effective timestamp.
Interface	interface_type, dataset, table/view/listing, API endpoint, event topic, sample queries	BigQuery table with Analytics Hub listing and authorized regional views.
Schema	field name, type, mode, description, sensitivity, policy tag, nullability, example	customer_id STRING REQUIRED, Confidential, policy tag: pii_identifier.
Quality	rules, thresholds, severity, owner, remediation path, exception process	customer_id non-null 100%; region_code reference-valid 99.9%.
SLO	freshness, availability, support response, incident severity, business calendar	95% refreshed within 15 minutes during business day.
Security	classification, access pattern, masking, row filters, export restriction, approval requirements	Restricted PII, masked for standard analysts, direct access prohibited.
Lifecycle	release policy, compatibility, deprecation window, retirement criteria	Minor versions compatible; major versions require 180-day migration window.
FinOps	cost center, labels, consumer attribution, chargeback rule, budget owner	business_domain=customer; product=customer_360; showback monthly.
Operations	dashboard URL, runbook	ServiceNow queue DATA-

Template section	Mandatory fields	Example
	URL, escalation, incident queue, maintenance window	CUST-PROD; 24x5 support.

3.3 Detailed RACI Matrix

Activity	Governance Council	Domain Owner	Data Steward	Product Manager	Product Engineer	Platform Team	Security	FinOps	Consumer
Approve product candidacy	C	A	R	R	I	I	C	C	C
Define business terms	A	A	R	C	I	I	C	I	C
Define product contract	C	A	R	R	C	C	C	C	C
Build pipeline and table	I	I	C	C	R	C	I	I	I
Define security controls	C	A	C	C	I	C	R	I	I
Approve restricted access	I	A	C	I	I	I	R	I	C
Publish marketplace listing	C	A	R	R	C	R	C	I	I
Monit	I	A	C	R	R	C	C	C	I

Activity	Governance Council	Domain Owner	Data Steward	Product Manager	Product Engineer	Platform Team	Security	FinOps	Consumer
or SLOs									
Remediate quality issue	I	A	R	C	R	C	I	I	C
Review product cost	I	A	I	R	C	C	I	R	C
Deprecate product	C	A	R	R	C	C	C	C	C
Approve exception	A	R	C	C	I	C	R	C	I

3.4 Data Product Taxonomy

Product type	Definition and design implication
Source-aligned product	A domain-owned product that exposes cleaned, documented and access-controlled data close to the source semantics. It is useful for teams that need detail and for building downstream products. It is not a raw dump; it includes schema, quality and lineage commitments.
Aggregate product	A product that combines several source-aligned products within one domain to answer recurring business questions. It usually has a more stable semantic model and narrower grain.
Consumer-aligned product	A product optimized for a particular consumer group or business process, such as finance close, demand planning or care-management operations. These products should avoid hard-coding one report when a reusable semantic product is possible.
Reference product	A stable set of enterprise reference values, hierarchies or mappings. It has stricter

Product type	Definition and design implication
	change control because many domains may depend on it.
Master-data product	A governed representation of core entities such as customer, product, supplier or employee. These products need explicit survivorship rules and stewardship.
Event product	A versioned event stream representing business changes, such as order created, shipment delayed or consent changed. It requires schema versioning, ordering and replay considerations.
AI-ready product	A certified product with additional metadata, allowed-use statements, sensitive-data controls and evaluation requirements for ML, RAG or agentic use cases.
External-sharing product	A product intended for partner, regulator or third-party consumption. It requires legal terms, minimization, residency and stronger audit evidence.

3.5 Domain Decomposition Workshop Outputs

Each domain decomposition workshop should produce artefacts that can be reviewed by architecture, governance and platform teams. These artefacts prevent disagreement later about ownership, scope and product priority.

Workshop artefact	Description	Review question
Capability map	Business capabilities grouped into candidate domains.	Does each candidate domain have a stable business owner?
Entity map	Core entities, identifiers and authoritative systems.	Which domain owns the meaning and lifecycle of each entity?
Source map	Applications, databases, SaaS platforms, files and streams feeding the domain.	Are sources grouped by semantics rather than platform convenience?
Consumer map	Teams, reports, applications, AI use cases and partners that need products.	Which consumers need certified products first?
Regulatory map	Sensitive data, residency, retention and legal constraints.	Which products require restricted design patterns?
Quality pain-point map	Known defects, reconciliation	Which defects block

Workshop artefact	Description	Review question
	issues and source inconsistencies.	certification and need remediation funding?
Product backlog	Prioritized candidate products with value and complexity.	Which first product proves the domain pattern end-to-end?
Dependency map	Upstream and downstream product dependencies.	Which cross-domain contracts must be negotiated?

3.6 Metadata and Catalog Standards

Metadata standards are the difference between a mesh that is discoverable and one that merely has many datasets. Each certified product must maintain metadata in four layers: technical metadata, business metadata, operational metadata and governance metadata. Automation should extract what systems know and require humans to add what only the business knows.

Metadata layer	Required fields	Source of truth
Technical	dataset, table, view, schema, partitioning, clustering, lineage, pipeline, code repository	Deployment metadata and Google Cloud asset metadata.
Business	description, business terms, grain, owner, steward, intended use, limitations	Contract repository and steward input.
Operational	freshness, quality score, incident history, SLO status, usage, downstream consumers	Monitoring, quality scans, logs and job metadata.
Governance	classification, policy tags, access rules, retention, legal hold, privacy deletion, certification status	Governance policy repository and catalog entries.

Catalog completeness criterion	Minimum standard for certification
Name and description	Human-readable product name and decision-oriented description.
Owner and steward	Named individuals or groups with support channel.
Classification	Product and field-level sensitivity recorded.
Glossary links	At least core business terms linked to enterprise glossary.
Quality status	Current quality score and rule summary

Catalog completeness criterion	Minimum standard for certification
	visible.
Freshness status	Last update and SLO status visible.
Lineage	Upstream sources and downstream dependencies captured or documented.
Access instructions	Consumer request process, approval requirements and usage obligations.
Examples	At least one safe sample query or access example.
Lifecycle status	Candidate, certified, deprecated or retired.

3.7 Data Mesh Metrics and KPIs

A data mesh must be measured as a product portfolio. Adoption alone is insufficient because broad use of low- quality data can increase enterprise risk. The portfolio dashboard should show value, trust, security, efficiency and operational health.

KPI category	Metric	Why it matters
Adoption	Active consumers per product	Shows reuse and product-market fit.
Reuse	Duplicate curated datasets retired	Measures whether the mesh reduces fragmentation.
Trust	Certified products as percentage of active products	Shows governance maturity.
Quality	Average critical rule pass rate	Indicates consumer reliability.
Freshness	Percentage of products meeting freshness SLO	Shows operational health.
Security	Restricted products with complete policy evidence	Measures security compliance.
Cost	Cost per certified product and cost per active consumer	Shows efficiency and chargeback readiness.
Lifecycle	Products deprecated or retired on schedule	Prevents stale marketplace assets.
Speed	Lead time from product approval to publication	Measures self-service platform effectiveness.
AI readiness	Products certified for AI use	Shows readiness for governed AI adoption.

3.8 Control Mapping for Regulated Data Products

Control objective	Mesh implementation	Evidence
Purpose limitation	Allowed-use statement in	Contract and workflow

Control objective	Mesh implementation	Evidence
	product contract and access request.	record.
Least privilege	Groups, authorized views, row filters and policy tags.	IAM export and policy configuration.
Data minimization	Consumer interface exposes only approved attributes.	View definition and approval record.
Access review	Periodic recertification for sensitive products.	Access review report.
Auditability	Query, access and publication logs retained.	Cloud Audit Logs and BigQuery audit views.
Retention	Retention policy defined at product and source layer.	Dataset/table/bucket policy evidence.
Privacy deletion	Deletion or suppression process defined for applicable entities.	Runbook and execution logs.
Legal hold	Products subject to hold cannot be deleted or modified outside policy.	Hold record and exception log.
Change control	Versioned contracts and CI/CD approvals.	Pull requests and deployment logs.
Incident response	Product runbook and escalation route.	Runbook and incident tickets.

3.9 Expanded Domain Product Catalogs

3.9.1 Healthcare Product Catalog

Product	Description	Classification	Primary sources	Consumers
Patient 360	One row per patient profile with identifiers, demographics, consent, care-team and risk flags.	PHI restricted	FHIR, EHR master patient index, consent platform	Clinical analytics, care management
Encounter Summary	Encounter-level clinical and administrative events.	PHI restricted	EHR, HL7/FHIR feeds	Operations, quality, regulatory
Claims Analytics	Claim header, line, adjudication and payment facts.	PHI and financial restricted	Claims platform, payer feeds	Finance, risk, actuarial

Product	Description	Classification	Primary sources	Consumers
Care Gap Registry	Preventive and chronic-care gap status by patient.	PHI restricted	EHR, quality rules, reference guidelines	Population health
Provider Network	Provider, facility and network participation data.	Confidential	Provider directory, credentialing	Access planning, network analytics

Product	Minimum quality gates	Standard SLO
Patient 360	Primary key uniqueness; freshness check; classification check; contract compatibility; reconciliation where applicable.	Tier based on criticality: 15 minutes to 24 hours freshness; 99.5%-99.9% availability.
Encounter Summary	Primary key uniqueness; freshness check; classification check; contract compatibility; reconciliation where applicable.	Tier based on criticality: 15 minutes to 24 hours freshness; 99.5%-99.9% availability.
Claims Analytics	Primary key uniqueness; freshness check; classification check; contract compatibility; reconciliation where applicable.	Tier based on criticality: 15 minutes to 24 hours freshness; 99.5%-99.9% availability.
Care Gap Registry	Primary key uniqueness; freshness check; classification check; contract compatibility; reconciliation where applicable.	Tier based on criticality: 15 minutes to 24 hours freshness; 99.5%-99.9% availability.
Provider Network	Primary key uniqueness; freshness check; classification check; contract compatibility; reconciliation where applicable.	Tier based on criticality: 15 minutes to 24 hours freshness; 99.5%-99.9% availability.

3.9.2 Finance Product Catalog

Product	Description	Classification	Primary sources	Consumers
General Ledger Gold	Certified account balances and journal facts by	Confidential/ restricted	ERP GL, subledger	Close, audit, reporting

Product	Description	Classification	Primary sources	Consumers
	legal entity and period.			
Profitability Cube	Revenue, cost and margin by product, customer and region.	Confidential	GL, sales, product, allocations	FP&A, executives
Treasury Liquidity	Cash position, forecast and bank account summaries.	Restricted	Treasury systems, bank feeds	Treasury
Tax Reporting Pack	Tax-relevant transactions and jurisdiction attributes.	Restricted	ERP, billing, procurement	Tax and compliance
Vendor Spend	Supplier and purchase spend with categories.	Confidential	Procurement, AP	Procurement, finance

Product	Minimum quality gates	Standard SLO
General Ledger Gold	Primary key uniqueness; freshness check; classification check; contract compatibility; reconciliation where applicable.	Tier based on criticality: 15 minutes to 24 hours freshness; 99.5%-99.9% availability.
Profitability Cube	Primary key uniqueness; freshness check; classification check; contract compatibility; reconciliation where applicable.	Tier based on criticality: 15 minutes to 24 hours freshness; 99.5%-99.9% availability.
Treasury Liquidity	Primary key uniqueness; freshness check; classification check; contract compatibility; reconciliation where applicable.	Tier based on criticality: 15 minutes to 24 hours freshness; 99.5%-99.9% availability.
Tax Reporting Pack	Primary key uniqueness; freshness check; classification check; contract compatibility; reconciliation where applicable.	Tier based on criticality: 15 minutes to 24 hours freshness; 99.5%-99.9% availability.
Vendor Spend	Primary key uniqueness; freshness check;	Tier based on criticality: 15 minutes to 24 hours

Product	Minimum quality gates	Standard SLO
	classification check; contract compatibility; reconciliation where applicable.	freshness; 99.5%-99.9% availability.

3.9.3 Human Resources Product Catalog

Product	Description	Classification	Primary sources	Consumers
Workforce Headcount	Current and historical worker population and status.	Restricted employee data	Workday, identity	Workforce planning
Organization Hierarchy	Manager and organization relationships.	Confidential/ restricted	HRIS, identity	Planning, access governance
Skills Inventory	Skills, certifications and learning completions.	Confidential	Learning systems, HRIS	Talent planning
Attrition Analytics	Employee movement and attrition features.	Restricted	HRIS, performance, engagement	HR analytics
Payroll Controls	Payroll reconciliation and control totals.	Restricted	Payroll, finance	Payroll operations, audit

Product	Minimum quality gates	Standard SLO
Workforce Headcount	Primary key uniqueness; freshness check; classification check; contract compatibility; reconciliation where applicable.	Tier based on criticality: 15 minutes to 24 hours freshness; 99.5%-99.9% availability.
Organization Hierarchy	Primary key uniqueness; freshness check; classification check; contract compatibility; reconciliation where applicable.	Tier based on criticality: 15 minutes to 24 hours freshness; 99.5%-99.9% availability.
Skills Inventory	Primary key uniqueness; freshness check; classification check; contract compatibility; reconciliation	Tier based on criticality: 15 minutes to 24 hours freshness; 99.5%-99.9% availability.

Product	Minimum quality gates	Standard SLO
	where applicable.	
Attrition Analytics	Primary key uniqueness; freshness check; classification check; contract compatibility; reconciliation where applicable.	Tier based on criticality: 15 minutes to 24 hours freshness; 99.5%-99.9% availability.
Payroll Controls	Primary key uniqueness; freshness check; classification check; contract compatibility; reconciliation where applicable.	Tier based on criticality: 15 minutes to 24 hours freshness; 99.5%-99.9% availability.

3.9.4 Sales Product Catalog

Product	Description	Classification	Primary sources	Consumers
Pipeline Forecast	Open opportunities, stages and forecast categories.	Confidential	CRM, forecast tools	Sales ops, executives
Account 360	Account hierarchy, activity, revenue and risk.	Confidential	CRM, billing, support	Sales, success
Bookings Gold	Certified bookings and order facts.	Confidential	CRM, CPQ, order management	Finance, revenue ops
Territory Assignment	Seller, account and territory coverage.	Confidential	Territory tools, CRM	Sales ops
Partner Performance	Partner pipeline, bookings and enablement indicators.	Confidential	Partner portal, CRM	Channel operations

Product	Minimum quality gates	Standard SLO
Pipeline Forecast	Primary key uniqueness; freshness check; classification check; contract compatibility; reconciliation where applicable.	Tier based on criticality: 15 minutes to 24 hours freshness; 99.5%-99.9% availability.
Account 360	Primary key uniqueness; freshness check;	Tier based on criticality: 15 minutes to 24 hours

Product	Minimum quality gates	Standard SLO
	classification check; contract compatibility; reconciliation where applicable.	freshness; 99.5%-99.9% availability.
Bookings Gold	Primary key uniqueness; freshness check; classification check; contract compatibility; reconciliation where applicable.	Tier based on criticality: 15 minutes to 24 hours freshness; 99.5%-99.9% availability.
Territory Assignment	Primary key uniqueness; freshness check; classification check; contract compatibility; reconciliation where applicable.	Tier based on criticality: 15 minutes to 24 hours freshness; 99.5%-99.9% availability.
Partner Performance	Primary key uniqueness; freshness check; classification check; contract compatibility; reconciliation where applicable.	Tier based on criticality: 15 minutes to 24 hours freshness; 99.5%-99.9% availability.

3.9.5 Marketing Product Catalog

Product	Description	Classification	Primary sources	Consumers
Campaign Performance	Campaign spend, engagement, conversion and ROI metrics.	Confidential	Marketing automation, ad platforms	Marketing analytics
Consent Status	Current consent and communication preferences.	Restricted PII	Consent platform, CRM	Marketing, privacy, sales
Audience Segments	Approved analytical and activation segments.	Restricted when identifiable	CDP, web analytics, CRM	Activation, analytics
Attribution Model	Multi-touch attribution metrics.	Confidential	Web analytics, CRM, campaign data	Marketing analytics
Lead Funnel	Lead lifecycle and conversion facts.	Confidential	Marketing automation, CRM	Marketing and sales ops

Product	Minimum quality gates	Standard SLO
Campaign Performance	Primary key uniqueness; freshness check; classification check; contract compatibility; reconciliation where applicable.	Tier based on criticality: 15 minutes to 24 hours freshness; 99.5%-99.9% availability.
Consent Status	Primary key uniqueness; freshness check; classification check; contract compatibility; reconciliation where applicable.	Tier based on criticality: 15 minutes to 24 hours freshness; 99.5%-99.9% availability.
Audience Segments	Primary key uniqueness; freshness check; classification check; contract compatibility; reconciliation where applicable.	Tier based on criticality: 15 minutes to 24 hours freshness; 99.5%-99.9% availability.
Attribution Model	Primary key uniqueness; freshness check; classification check; contract compatibility; reconciliation where applicable.	Tier based on criticality: 15 minutes to 24 hours freshness; 99.5%-99.9% availability.
Lead Funnel	Primary key uniqueness; freshness check; classification check; contract compatibility; reconciliation where applicable.	Tier based on criticality: 15 minutes to 24 hours freshness; 99.5%-99.9% availability.

3.9.6 Supply Chain Product Catalog

Product	Description	Classification	Primary sources	Consumers
Inventory Position	SKU/location inventory, availability and movements.	Confidential	WMS, ERP, IoT	Operations, planning
Shipment Visibility	Shipment events, carrier status and delay predictions.	Confidential	TMS, carrier APIs, IoT	Logistics, customer service
Supplier Risk	Supplier performance, disruptions and risk indicators.	Confidential/ restricted	Supplier portal, risk feeds	Procurement, resilience
Demand Forecast Inputs	Historical demand and external drivers	Confidential	Orders, sales, external signals	Planning, AI

Product	Description	Classification	Primary sources	Consumers
	for forecasting.			
Order Fulfillment	Order status, pick/pack/ship and exception metrics.	Confidential	OMS, WMS, TMS	Operations, sales

Product	Minimum quality gates	Standard SLO
Inventory Position	Primary key uniqueness; freshness check; classification check; contract compatibility; reconciliation where applicable.	Tier based on criticality: 15 minutes to 24 hours freshness; 99.5%-99.9% availability.
Shipment Visibility	Primary key uniqueness; freshness check; classification check; contract compatibility; reconciliation where applicable.	Tier based on criticality: 15 minutes to 24 hours freshness; 99.5%-99.9% availability.
Supplier Risk	Primary key uniqueness; freshness check; classification check; contract compatibility; reconciliation where applicable.	Tier based on criticality: 15 minutes to 24 hours freshness; 99.5%-99.9% availability.
Demand Forecast Inputs	Primary key uniqueness; freshness check; classification check; contract compatibility; reconciliation where applicable.	Tier based on criticality: 15 minutes to 24 hours freshness; 99.5%-99.9% availability.
Order Fulfillment	Primary key uniqueness; freshness check; classification check; contract compatibility; reconciliation where applicable.	Tier based on criticality: 15 minutes to 24 hours freshness; 99.5%-99.9% availability.

3.10 Common Anti-Patterns and Corrective Actions

Anti-pattern	Why it fails	Correction
Every application becomes a domain	Creates technical silos instead of business ownership.	Decompose by business capability and semantic ownership.
Marketplace without certification	Consumers cannot determine what is trusted.	Use certification levels and quality evidence.
Raw data published as	Consumers inherit source	Require contract, quality,

Anti-pattern	Why it fails	Correction
product	complexity and quality defects.	documentation and support.
Central team owns all products	Delivery becomes a bottleneck and semantics remain detached from business owners.	Shift ownership to domains with platform support.
Policy documents without automation	Controls are inconsistent and hard to audit.	Implement computational governance through CI/CD and metadata.
Broad access for convenience	Creates privacy and regulatory risk.	Use authorized views, policy tags, row filters and request workflows.
No deprecation process	Consumers break or continue using stale data.	Require versioning and retirement gates.
Cost ignored until late	Domain autonomy becomes cost sprawl.	Mandate product labels and cost dashboards from day one.
AI allowed by default	Sensitive data can be misused by models or agents.	Create AI-ready certification and explicit allowed-use metadata.

3.11 Review Agendas

Review	Participants	Agenda
Domain onboarding review	Domain owner, steward, platform, security, FinOps	Domain boundaries, sources, product backlog, project structure, classification and first pilot.
Product design review	Product manager, steward, engineers, security, consumers	Contract, interface, SLOs, access, quality rules, cost and lifecycle.
Certification review	Owner, steward, governance delegate, platform, security	Evidence, exceptions, quality, lineage, catalog and marketplace readiness.
Operational readiness review	Product team, SRE, platform, consumers	Dashboards, alerts, runbooks, escalation, recovery, cost and support.
Quarterly product review	Domain leadership, consumers, FinOps, governance	Usage, value, quality, cost, incidents, roadmap and retirement candidates.

3.12 Production Acceptance Criteria

Category	Acceptance criterion
Architecture	Product aligns to approved domain and interface pattern.
Security	Classification, policy tags, row policies, masking and access workflow are configured as required.
Governance	Contract, catalog entry, owner, steward and glossary mapping are complete.
Quality	Critical rules pass and scorecard is visible.
Reliability	Freshness and availability monitoring exists.
Operations	Runbook, escalation and incident severity mapping exist.
FinOps	Labels, cost center and dashboard are available.
Marketplace	Listing or access documentation is complete.
Consumer adoption	At least one named consumer has validated the product.
Lifecycle	Versioning and deprecation policy is documented.